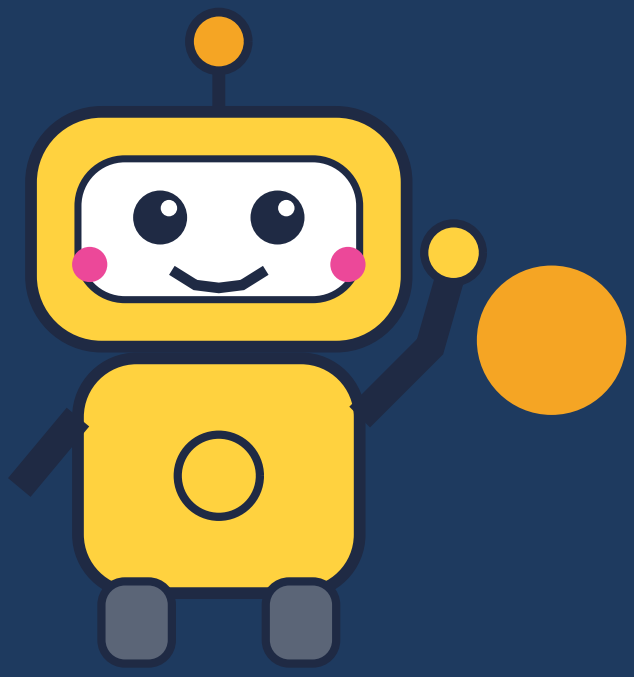
SAMPLE PAPERS

Think Like a Coder

Sample examination papers, Class 6

Half-yearly and annual papers
50 marks, 2 hours, with blueprints and marking schemes
Practical stations and project assessment



For CBSE's Computational Thinking and AI curriculum, 2026-27

Goes with Think Like a Coder, Class 6 · Photocopiable for school use

SAMPLE EDITION · DRAFT

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About these papers

These are two sample examination papers for Class 6: a **half-yearly** paper on the term-1 chapters and an **annual** paper on the whole year. They go with *Think Like a Coder*, Class 6, and with the chapter assessments. All questions are new.

How they follow CBSE

- CBSE does not hold board examinations for Classes 3-8 and has not published a question-paper pattern for CT&AI.; Schools set their own papers. These papers follow what CBSE's CT&AI; handbooks describe: **competency-based** written tests with CT puzzles, with no 'write a program' questions and no computer needed.
- The question types are the ones CBSE uses in its competency-based papers: multiple choice, assertion-reason, short and long answers, and a case-based question.
- Questions are spread across four thinking levels (remember, understand, apply, analyse and create), with most marks for applying and analysing. The blueprints show the split.
- Chapters follow the order of CBSE's 2026-27 CT&AI; handbooks and the NCERT maths chapters, including the four AI chapters.
- The handbooks also assess Classes 6-8 through practical examinations, project presentations and reflective journals. The last section gives unplugged practical stations, a rubric and one way to combine the marks.

Paper design

Section	Question type	Questions	Marks
A	Multiple choice and assertion-reason	10	10 x 1 = 10
B	Very short answer	5	5 x 2 = 10
C	Short answer	5	5 x 3 = 15
D	Long answer	2	2 x 5 = 10
E	Case-based question	1	1 x 5 = 5
	Total · 2 hours	23	50

Using the papers

- Use the half-yearly paper in September and the annual paper in February-March, or as mock tests before the school's own exams.
- Print one paper per student; students answer on separate answer sheets.
- Read questions aloud to students who find reading hard: the paper tests thinking, not reading.
- Mark with the marking scheme; accept any correct method, and give part marks for correct steps.
- Use the blueprints to set your own papers in the same pattern.

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SAMPLE PAPER · HALF-YEARLY EXAMINATION

Class 6 · Computational Thinking and Artificial Intelligence

Syllabus: chapters 1-5 and AI 1-2 (term 1)

Time allowed: 2 hours **Maximum marks:** 50

Name: _____ **Section:** _____ **Roll no.:** _____

General instructions

1. This paper has 23 questions in 5 sections. All questions are compulsory.
2. Section A: multiple choice and assertion-reason, 10 questions of 1 mark each; Section B: very short answer, 5 questions of 2 marks each; Section C: short answer, 5 questions of 3 marks each; Section D: long answer, 2 questions of 5 marks each; Section E: case-based question, 1 question of 5 marks each.
3. Marks for each question are shown on the right.
4. Write your answers on the separate answer sheet. Draw tables and diagrams neatly where asked.
5. No calculator or computer is needed. Show your working: marks are given for correct steps.

SECTION A · Multiple choice and assertion-reason — 10 x 1 = 10 marks

1. What is the next term? 2, 6, 10, 14, ... [1]

a) 16 **b)** 18 **c)** 20

2. A function takes an input and gives back ... [1]

a) an output **b)** an error **c)** a loop

3. An angle between 90° and 180° is ... [1]

a) acute **b)** obtuse **c)** reflex

4. Which number is a palindrome? [1]

a) 1221 **b)** 1212 **c)** 1122

5. In the list [7, 3, 9, 5], the biggest item is at position ... [1]

a) 1 **b)** 3 **c)** 4

6. Which number is neither prime nor composite? [1]

a) 1 **b)** 2 **c)** 9

7. A washing machine that runs the same fixed cycle every time is an example of ... [1]

a) AI **b)** automation

8. Data that has been organised to answer a question is called ... [1]

a) information **b)** noise **c)** a password

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9. **Assertion (A):** 51 is not a prime number. [1]
Reason (R): 51 = 3 x 17, so it has more than two factors.

a) Both A and R are true, and R explains A.

b) Both A and R are true, but R does not explain A.

c) A is true, but R is false.

d) A is false, but R is true.

10. **Assertion (A):** A spam filter is an example of AI. [1]
Reason (R): A spam filter follows the same fixed rule every time and never changes.

a) Both A and R are true, and R explains A.

b) Both A and R are true, but R does not explain A.

c) A is true, but R is false.

d) A is false, but R is true.

SECTION B · Very short answer — 5 x 2 = 10 marks

11. triangle(n) = 1 + 2 + ... + n. Find triangle(6) and triangle(10). [2]

12. A robot faces East on an 8-point compass. It turns 135° anticlockwise. Which way does it face now? [2]

13. Reverse and add until you get a palindrome, starting with 39. Show each step. [2]

14. List the common factors of 18 and 24. Which is the highest? [2]

15. Name two ways of collecting data, with an example of each. [2]

SECTION C · Short answer — 5 x 3 = 15 marks

16. Write a function rectangle(l, w) that draws a rectangle with turns. Write the call that draws a rectangle 60 long and 30 wide. What do its turns add up to? [3]

17. Write an algorithm to find the smallest item in a list. Trace it on 12, 7, 15, 4, 9. [3]

18. List the prime numbers between 30 and 50. Say which multiples you cross out to find them. [3]

19. Repeat until n = 1: if n is even, n = n ÷ 2, else n = 3n + 1. Start with n = 7. Write every value and count the steps. [3]

20. Explain supervised, unsupervised and reinforcement learning, with one example of each. [3]

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SECTION D · Long answer — 2 x 5 = 10 marks

21. A row of houses is made with sticks: 1 house uses 5 sticks, 2 houses use 9 and 3 houses use 13.

(a) Write a 'next term' rule. [1]

(b) Write a 'position' rule for n houses. [2]

(c) How many sticks do 25 houses need? [1]

(d) How many houses can be made with 81 sticks? [1]

22. Plan a survey to answer: 'How much water do students in our school drink in a day?'

(a) Write your survey question and its answer options. [1]

(b) Say whom you will ask and how. [1]

(c) Draw the table you will use to organise the answers. [1]

(d) Which chart will you use to show the results? Why? [1]

(e) Give one way to keep the data private and safe. [1]

SECTION E · Case-based question — 1 x 5 = 5 marks

23. Rainfall (in mm) for one week is stored as a list: Mon 12, Tue 0, Wed 25, Thu 8, Fri 30, Sat 5, Sun 0.

(a) What is at position 5 of the list? [1]

(b) Which day had the most rain and which had the least? [1]

(c) What was the total rainfall? [1]

(d) In a pictograph, 1 symbol = 5 mm. How many symbols show Wednesday? [1]

(e) On which days was there more than 10 mm of rain? [1]

***** End of the question paper *****

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SAMPLE PAPER · ANNUAL EXAMINATION

Class 6 · Computational Thinking and Artificial Intelligence

Syllabus: all chapters, mostly chapters 6-10 and AI 3-4 (term 2)

Time allowed: 2 hours **Maximum marks:** 50

Name: _____ **Section:** _____ **Roll no.:** _____

General instructions

1. This paper has 23 questions in 5 sections. All questions are compulsory.
2. Section A: multiple choice and assertion-reason, 10 questions of 1 mark each; Section B: very short answer, 5 questions of 2 marks each; Section C: short answer, 5 questions of 3 marks each; Section D: long answer, 2 questions of 5 marks each; Section E: case-based question, 1 question of 5 marks each.
3. Marks for each question are shown on the right.
4. Write your answers on the separate answer sheet. Draw tables and diagrams neatly where asked.
5. No calculator or computer is needed. Show your working: marks are given for correct steps.

SECTION A · Multiple choice and assertion-reason — 10 x 1 = 10 marks

1. area(9, 4) = ... [1]

a) 13 **b)** 26 **c)** 36

2. Which fraction is in its simplest form? [1]

a) 6/9 **b)** 5/8 **c)** 4/10

3. In a construction, the compass is used to ... [1]

a) draw arcs and circles **b)** measure angles

c) colour shapes

4. What is the order of rotational symmetry of a regular pentagon? [1]

a) 1 **b)** 5 **c)** 10

5. -3 - 4 = ... [1]

a) -7 **b)** -1 **c)** 7

6. A rule learnt from labelled examples is used to ... [1]

a) predict labels for new examples **b)** delete data

c) draw charts

7. Copying someone's essay and calling it your own is ... [1]

a) phishing **b)** plagiarism **c)** piracy

8. A robot draws a regular hexagon. Each turn is ... [1]

a) 45° **b)** 60° **c)** 120°

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9. **Assertion (A):** -8 is smaller than -2. [1]
Reason (R): On a number line, -8 is to the left of -2.

a) Both A and R are true, and R explains A.

b) Both A and R are true, but R does not explain A.

c) A is true, but R is false.

d) A is false, but R is true.

10. **Assertion (A):** The password 'Rohan2014' is strong. [1]
Reason (R): Strong passwords are long and mix words, numbers and symbols that are hard to guess.

a) Both A and R are true, and R explains A.

b) Both A and R are true, but R does not explain A.

c) A is true, but R is false.

d) A is false, but R is true.

SECTION B · Very short answer — 5 x 2 = 10 marks

11. Find the perimeter and the area of a rectangle 12 m long and 5 m wide. [2]

12. Write three fractions equal to 2/5. [2]

13. Columns are numbered 1 to 10, with a mirror line between 5 and 6. Where do squares in column 2 and column 5 go when reflected? [2]

14. Temperatures were -6°C, 4°C, -1°C and 9°C. Find the range. [2]

15. Observation or conclusion? (a) 18 of the 20 apples in this basket are red. (b) Apples are usually red. [2]

SECTION C · Short answer — 5 x 3 = 15 marks

16. A plot is a 10 m by 8 m rectangle with a 4 m by 3 m rectangle cut out of one corner. Find its area and its perimeter. [3]

17. Sort 2/3, 1/2, 3/4 and 5/12 from smallest to biggest with bubble sort. First write them in twelfths; then show each pass. [3]

18. A script to find the midpoint of AB = 8 cm says: 1. Draw AB. 2. With the compass at A and radius 3 cm, draw arcs above and below AB. 3. With the compass at B and radius 5 cm, draw arcs. 4. Join the points where the arcs cross. Find two bugs and fix the script. [3]

19. A message says: 'Congratulations! You have won a phone. Send your parent's bank OTP within 10 minutes to claim it.' Give three signs that this is phishing. [3]

20. Labelled examples for 'Does it float?': wooden block, floats; iron nail, sinks; plastic ball, floats; steel spoon, sinks; cork, floats. Find a rule. Use it to predict a glass marble. Is the prediction right? [3]

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SECTION D · Long answer — 2 x 5 = 10 marks

21. A lift serves floors -3 to 10. It starts at floor 0 and makes these moves: +5, -7, +2, -1, +6.

(a) Trace the floor after each move. [2]

(b) What is the lowest floor reached? [1]

(c) How many floors does the lift travel in all? [1]

(d) Write a rule that stops the lift going below floor -3. [1]

22. Symmetry in letters and shapes.

(a) Which letters of the word MATH have a line of symmetry going straight down? [1]

(b) Which letters of the word NOSE have rotational symmetry of order 2? [1]

(c) Name a shape with exactly 2 lines of symmetry and rotational symmetry of order 2. [1]

(d) Columns are numbered 1 to 8, with a mirror between 4 and 5. Write a rule for the new column after reflecting. [1]

(e) Explain the difference between line symmetry and rotational symmetry. [1]

SECTION E · Case-based question — 1 x 5 = 5 marks

23. A school wants an AI app that predicts which students may need extra help in maths, using last year's marks and attendance.

(a) What labelled examples would it learn from? [1]

(b) Why should it be tested on new students' data before it is used? [1]

(c) Give one privacy risk. [1]

(d) Who should make the final decision about extra help? Why? [1]

(e) Give one way bias could get into the app. [1]

***** End of the question paper *****

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Blueprints

Blueprint: Half-yearly examination

Syllabus: chapters 1-5 and AI 1-2 (term 1). Marks by chapter and section.

Chapter	A	B	C	D	E	Marks
1. Patterns in Mathematics	2	2		5		9
2. Lines and Angles	1	2	3			6
3. Number Play	1	2	3			6
4. Data Handling and Presentation	1		3		5	9
5. Prime Time	2	2	3			7
AI 1. Introduction to AI	2		3			5
AI 2. Basic Data Concepts	1	2		5		8
Total	10	10	15	10	5	50

Marks by thinking level

Each question is counted at its main thinking level.

Level	What the student does	Marks	Share
Remember	recall a fact or term	4	8%
Understand	explain or interpret an idea	8	16%
Apply	use an idea or rule in a new problem	26	52%
Analyse and create	trace, debug, compare, design or justify	12	24%

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Blueprint: Annual examination

Syllabus: all chapters, mostly chapters 6-10 and AI 3-4 (term 2). Marks by chapter and section.

Chapter	A	B	C	D	E	Marks
2. Lines and Angles	1					1
6. Perimeter and Area	1	2	3			6
7. Fractions	1	2	3			6
8. Playing with Constructions	1		3			4
9. Symmetry	1	2		5		8
10. The Other Side of Zero	2	2		5		9
AI 3. Patterns and Decisions	1	2	3		5	11
AI 4. Ethics and Digital Responsibility	2		3			5
Total	10	10	15	10	5	50

Marks by thinking level

Each question is counted at its main thinking level.

Level	What the student does	Marks	Share
Remember	recall a fact or term	2	4%
Understand	explain or interpret an idea	7	14%
Apply	use an idea or rule in a new problem	24	48%
Analyse and create	trace, debug, compare, design or justify	17	34%

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Marking schemes

Accept any correct method and any sensible wording. Give part marks for correct steps, even when the final answer has a slip.

Marking scheme: Half-yearly examination

1. b) 18
2. a) an output
3. b) obtuse
4. a) 1221
5. b) 3
6. a) 1
7. b) automation
8. a) information
9. a) Both A and R are true, and R explains A.
10. c) A is true, but R is false. (Spam filters learn from labelled examples.)
11. 21 and 55 (1 mark for each)
12. North-West (2 marks; 1 mark for counting 3 steps of 45°)
13. 39 + 93 = 132; 132 + 231 = 363 (1 mark for each step)
14. 1, 2, 3, 6; the highest is 6 (1 mark for the list, 1 mark for 6)
15. For example: a survey (asking classmates their favourite sport); observation (counting vehicles at the gate); measuring (students' heights) (1 mark for each way with an example)
16. define rectangle(l, w): repeat 2 times: forward l, turn right 90°, forward w, turn right 90°. Call: rectangle(60, 30). The turns add up to 360°. (1 mark for the function, 1 mark for the call, 1 mark for 360°)
17. Set smallest to the first item; for each item, if item < smallest then smallest = item. Trace: 12, 7, 7, 4, 4. Smallest is 4. (1 mark for the algorithm, 1 mark for the trace, 1 mark for 4)
18. 31, 37, 41, 43, 47. Cross out multiples of 2, 3, 5 and 7 (35 and 49 are the multiples of 7 left). (2 marks for the list (1 mark with one missing or extra), 1 mark for the explanation)
19. 7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1: 16 steps (2 marks for the values (1 mark with one slip), 1 mark for 16)
20. Supervised: learns from labelled examples, such as photos labelled cat or dog. Unsupervised: finds groups with no labels, such as grouping customers. Reinforcement: learns by trial, reward and penalty, such as a game-playing robot. (1 mark for each)
21. (a) add 4 (b) sticks = 4n + 1 (c) 101 (d) 20 houses ((a) 1 mark (b) 2 marks (c) 1 mark (d) 1 mark)
22. Open answer, for example: (a) 'How many glasses of water did you drink yesterday?' with options 0-2, 3-5, 6-8, more than 8 (b) ask 10 students from each class (c) a frequency table of options and counts (d) a bar chart, because it compares groups (e) do not write names; store the sheets safely (1 mark for each part)
23. (a) 30 (b) highest: Friday, 30 mm; lowest: Tuesday and Sunday, 0 mm (c) 80 mm (d) 5 symbols (e) Monday, Wednesday and Friday (1 mark for each part)

Marking scheme: Annual examination

1. c) 36
2. b) 5/8
3. a) draw arcs and circles
4. b) 5
5. a) -7
6. a) predict labels for new examples
7. b) plagiarism
8. b) 60°
9. a) Both A and R are true, and R explains A.
10. d) A is false, but R is true. (A name and a year are easy to guess.)
11. 34 m; 60 m^2 (1 mark for each)
12. For example 4/10, 6/15, 8/20 (2 marks for three; 1 mark for two)
13. column 9 and column 6 (1 mark for each)
14. 9 - (-6) = 15°C (1 mark for the method, 1 mark for 15)
15. (a) observation (b) conclusion (1 mark for each)
16. Area: 80 - 12 = 68 m^2. Perimeter: 36 m (cutting a corner out does not change the distance around). (1 mark for the method, 1 mark for 68 m^2, 1 mark for 36 m)
17. Twelfths: 8, 6, 9, 5. Pass 1: 6, 8, 5, 9. Pass 2: 6, 5, 8, 9. Pass 3: 5, 6, 8, 9. So 5/12, 1/2, 2/3, 3/4. (1 mark for twelfths, 1 mark for the passes, 1 mark for the final order)
18. Bug 1: 3 cm is not more than half of AB (4 cm), so the arcs may not meet. Bug 2: the radius must be the same from A and from B. Fix: use 5 cm from both A and B. (1 mark for each bug, 1 mark for the fix)
19. It is too good to be true; it asks for an OTP (secret bank information); it rushes you; it comes from an unknown sender. (1 mark for each sign, up to 3)
20. Rule: if it is made of metal, it sinks; else it floats. Prediction: the marble floats. Wrong: a glass marble sinks, so the rule needs more varied examples. (1 mark for the rule, 1 mark for the prediction, 1 mark for spotting the error)
21. (a) 5, -2, 0, -1, 5 (b) -2 (c) 21 floors (d) If floor + move < -3 then show 'Not allowed', else move. ((a) 2 marks, 1 mark with one slip (b) 1 mark (c) 1 mark (d) 1 mark)
22. (a) M, A, T and H (b) N, O and S (c) for example a rectangle (d) new column = 9 - column (e) Line symmetry: folding makes the halves match. Rotational symmetry: turning by less than a full turn looks the same. (1 mark for each part)
23. (a) Last year's students' marks and attendance, each labelled 'needed help' or 'did not' (b) to check it works on data it has not seen (c) for example, students' marks could be seen by others or shared without permission (d) the teacher, because people are responsible for decisions (e) for example, if the data came only from one section, it may be unfair to others (1 mark for each part)

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Practical and project assessment

CBSE's Classes 6-8 handbooks assess CT&AI; through practical examinations, project presentations and reflective journals as well as written tests. CBSE has not fixed how the marks are split, so the school decides. One way to make a 100-mark annual result:

Component	Marks	Evidence
Written examination (this paper)	50	Annual sample paper
Practical examination (unplugged)	20	Two stations from the list below, 10 marks each
Projects (two 20-hour projects)	20	Project rubrics in the workbook; presentation and data sheets
Observation journal and reflection	10	Teacher Observation Journal levels from the teacher book

Practical examination stations

Each station takes about 15 minutes. Students work alone or in pairs with paper, cards or the classroom kit; the examiner watches, asks the two questions shown, and marks with the rubric. No computer is needed.

Station	Task	Examiner asks
1. Function machine (chapter 1)	Find the hidden rule from four input-output cards, then predict the output for a new input.	How did you find the rule? How can you check it?
2. Drawing robot (chapter 2)	Write forward-and-turn steps for a partner to draw a regular hexagon on grid paper.	Why is each turn 60°? What do the turns add up to?
3. The sieve (chapter 5)	Use the 1-50 sieve card to find all the primes.	Why can you stop after crossing out multiples of 7? Is 1 prime?
4. Bubble sort (chapter 7)	Sort six fraction cards from smallest to biggest by swapping neighbours; count the swaps.	How do you compare two fractions? How do you know the list is sorted?
5. AI sorter (AI 3)	Learn a rule from eight labelled picture cards, then label four new cards.	Which feature decides the label? What would make the rule better?
6. Stay safe online (AI 4)	Rate five passwords as weak or strong, and sort three messages into safe and phishing.	What makes a password strong? What would you do with the phishing message?

Rubric for each station (10 marks)

Criterion	Marks	Full marks when the student ...
Understands the task	2	restates the goal and what a correct result looks like
Method	4	follows or designs clear, ordered steps; uses the right CT or AI idea
Result and checking	2	gets a correct result and checks or tests it
Explaining	2	answers the examiner's questions with reasons, using the right words

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